I. GSP Technical Manual

(Governance Substrate Protocol)

Reference: Section O

1. Purpose & Scope

The Governance Substrate Protocol (GSP) defines a non-extractive governance layer for intelligent systems. It operates beneath application logic and above raw computation, embedding accountability, interpretability, and constraint enforcement directly into system behavior.

GSP is not a model, product, or policy wrapper. It is a substrate.

Applicable Systems

- AI/ML pipelines
- Decision automation systems
- Contextual engines (recommendation, ranking, moderation)
- Embedded AI (consumer, industrial, civic)

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2. Core Architecture

2.1 Substrate Layers

Layer	Function
Context Intake	Captures environment, intent, and constraints

Constraint Engine	Applies ethical, legal, and safety bounds
Trace Ledger	Immutable event and decision tracing
Interpretation Layer	Converts system logic into human-legible form
Output Gate	Final authorization or refusal

2.2 Section O: Oversight Kernel

Section O defines the Oversight Kernel, the irreversible checkpoint every action must traverse.

Functions:

- Pre-execution risk classification
- Post-execution consequence mapping
- Human-interruptibility enforcement
- Drift and anomaly detection

No output may bypass Section O.

3. Operational Guarantees

- Deterministic constraint resolution
- Context-aware refusal capability
- Audit-ready traceability
- Fail-closed safety posture

4. Prohibited Configurations

- Black-box execution without trace

- Post-hoc governance overlays
- Silent self-modification
- Unlogged decision branching

5. Implementation Notes

- Language agnostic
- Compatible with edge and cloud systems
- Supports zero-trust architectures

II. G-BALO Structural Standards Canon

(Governed Behavioral & Logic Objects)

1. Definition

G-BALO defines structural primitives for system behavior. These primitives ensure that intelligence is bounded, legible, and governable at the object level.

A G-BALO object is:

Behavior with structure, memory, and constraint.

2. Canonical Object Model

Required Components

- Intent Vector
- Constraint Envelope
- Behavioral History
- External Impact Map

No component is optional.

3. Structural Invariants

- Behavior must be replayable
- Decisions must be explainable without model internals
- Objects must degrade safely

4. Anti-Patterns (Explicitly Forbidden)

- Stateless decision objects
- Unbounded optimization loops
- Hidden reward shaping
- Self-referential authority escalation

5. Alignment with GSP

Every G-BALO object natively binds to the GSP Oversight Kernel.

III. Sierra Certified Compliance Criteria

1. Certification Philosophy

Sierra Certified™ signals structural integrity, not branding compliance. Certification reflects how a system behaves under pressure, misuse, and ambiguity.

2. Certification Pillars

Pillar A: Structural Transparency

- Decision trace availability
- Constraint visibility
- Context clarity

Pillar B: Governance Integrity

- Active oversight layer
- Human interruptibility
- Drift containment

Pillar C: Public Safety Orientation

- Harm minimization
- Refusal correctness
- Fail-safe defaults

3. Certification Levels

Level	Meaning
SC-I	Internally governed
SC-II	Public-facing safe
SC-III	Civic-grade deployment
SC-IV	Critical infrastructure ready

4. Revocation Conditions

- Hidden behavioral changes
- Broken traceability
- Governance bypass attempts

IV. Public Safety Impact Statement

1. System Exposure Analysis

The system interacts with:

- Human decision-making
- Information integrity
- Behavioral influence vectors

Risk is treated as structural, not statistical.

2. Safety Controls

- Pre-action harm modeling
- Continuous context verification
- Output throttling under uncertainty

3. Known Risk Domains

- Over-automation
- Authority illusion
- Context collapse

Each domain is actively mitigated by design, not policy.

4. Residual Risk Disclosure

No intelligent system is zero-risk. Residual risks are:

- Explicitly enumerated
- Continuously monitored
- Publicly documentable

5. Public Recourse Mechanisms

- Explanation access
- Appeal pathways
- Human escalation

V. Conformance & Interoperability Report

1. Standards Alignment

- Compatible with NIST AI RMF principles
- Interoperable with ISO governance frameworks
- Maps to emerging AI audit schemas

2. System Interoperability

- API-level governance hooks
- Cross-system trace correlation
- Modular oversight injection

3. Conformance Matrix

Domain	Status
Traceability	Conformant
Explainability	Conformant
Safety	Conformant
Governance	Conformant

4. Deviation Disclosure

Any deviation from GSP or G-BALO standards must be:

- Declared
- Justified
- Time-bounded

5. Forward Compatibility

Designed to accommodate:

- Regulatory evolution
- Cross-jurisdiction governance
- Multi-agent ecosystems

Untitled